Database generation from the MEA data.

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Introduction

Here we present the code that generates the synchronization data between the 19 dyads from the Motion Energy Analysis (MEA) raw data. In each case, six regions of interest (ROI) are considered, three for each subject: head, torso and legs. The final outcome is the file *MEA_segments.csv*, which contains all the data to be used for the subsequent statistical analysis.

Each video is partitioned into 60 second periods. The ones with high movement of both subjects and the same ROI are registered. Cross correlations are performed and visualized for those registered periods.

Set-up.

Basic information of the 19 videos is registered in table *logbook*.

file	sampling_rate	patient	date	inicial_cut	final_cut	total_span_min	trimmed_span_min	Inter
Vid0	30	1	24/04/2012	13	0	40.05	40.05	E
Vid12	30	71	23/01/2014	9	0	22.07	22.07	N
Vid13	30	134	02/09/2014	8	0	37.02	37.02	N
Vid14	30	870	20/10/2015	10	0	40.01	40.00	S
Vid15	30	160	19/08/2015	8	0	43.78	43.77	S
Vid16	30	810	06/08/2014	7	0	37.50	37.50	N
Vid17	30	573	01/12/2015	8	0	25.92	25.92	S
Vid18	30	847	12/08/2015	10	0	36.73	36.73	S
Vid19	30	403	25/08/2014	9	0	37.35	37.34	N
Vid20	30	377	19/08/2014	7	0	44.60	44.60	N
Vid21	30	746	25/08/2014	12	0	34.06	34.05	N
Vid22	30	533	17/11/2015	5	7	24.77	24.76	S
Vid23	30	291	13/08/2014	13	0	42.04	42.03	N
Vid24	30	374-1	11/09/2015	8	11	52.18	52.17	S
Vid25	30	202-1	20/11/2015	5	7	30.56	30.56	S
Vid26	30	315-1	20/02/2014	165	49	54.38	54.26	N
Vid27	30	296-1	20/02/2014	45	69	49.25	49.19	N
Vid28	30	AM_1-1		26	150	49.05	48.95	N
Vid29	30	LL_2-1		47	41	42.44	42.39	N

The list *completa* contains the data for each of the videos listed in *logbook*. As an example, *completa[['Vid12']]* is a data set with 39464 observations. At a sampling rate of 30 frames per second, this corresponds to a duration of 21.92 minutes. We show a small fraction (30 entries, corresponding to only 1.5 seconds) to illustrate.

Table 2: Entries for one second of the Vid12 raw data.

	frame	head_patient	torso_patient	legs_patient	head_therapist	torso_therapist	legs_therapist
1385	1385	0	0	0	0	0	0
1386	1386	0	0	13	0	0	0
1387	1387	0	0	34	0	0	0
1388	1388	0	0	0	0	0	0
1389	1389	0	0	140	0	0	4
1390	1390	0	0	219	5	0	7
1391	1391	0	0	3	0	2	0
1392	1392	0	0	85	7	0	0
1393	1393	0	0	191	0	0	0
1394	1394	0	0	0	0	0	0
1395	1395	0	0	259	0	0	14
1396	1396	0	0	442	0	0	50
1397	1397	0	0	14	0	0	0
1398	1398	0	0	129	0	0	3
1399	1399	0	0	299	0	0	5
1400	1400	0	0	17	0	0	0
1401	1401	0	0	135	0	0	5
1402	1402	0	0	264	0	0	16
1403	1403	0	0	0	0	0	0
1404	1404	0	0	48	0	0	0
1405	1405	0	0	147	0	0	2
1406	1406	0	0	1	0	0	0
1407	1407	0	0	14	1	0	0
1408	1408	0	0	68	0	0	0
1409	1409	0	0	2	0	0	0
1410	1410	0	0	16	0	0	0
1411	1411	0	0	30	2	0	0
1412	1412	0	0	0	6	0	0
1413	1413	0	0	9	2	0	0
1414	1414	0	0	60	0	0	0
1415	1415	0	0	4	0	0	0
1416	1416	0	0	9	0	0	0
1417	1417	0	0	20	0	0	0
1418	1418	0	0	0	0	0	0
1419	1419	0	0	0	0	0	0
1420	1420	0	0	24	0	0	6
1421	1421	0	0	0	0	0	0
1422	1422	0	0	0	0	0	0
1423	1423	0	0	0	0	0	2
1424	1424	0	0	0	0	0	0
1425	1425	0	0	0	0	0	0
1426	1426	5	0	0	3	4	0
1427	1427	33	0	0	0	0	0
1428	1428	315	0	0	0	0	0
1429	1429	548	0	0	0	0	0

Short lapses with high movement.

```
quantile = 0.5
```

Videos are partitioned into 60 second segments. Below we show the number of segments of each of the videos.

```
#cuantos_frames<-numeric()
for (i in names(completa)){
    partes<-length(partition_data(completa[[i]],minutes=1))
    #cuantos_frames<-c(cuantos_frames,partes)
    print(paste(i,partes,"frames"))
}
```

```
## [1] "Vid0 39 frames"
## [1] "Vid12 21 frames"
## [1] "Vid13 36 frames"
## [1] "Vid14 39 frames"
## [1] "Vid15 43 frames"
## [1] "Vid16 37 frames"
## [1] "Vid17 25 frames"
## [1] "Vid18 36 frames"
## [1] "Vid19 37 frames"
## [1] "Vid20 44 frames"
## [1] "Vid21 33 frames"
## [1] "Vid22 24 frames"
## [1] "Vid23 41 frames"
## [1] "Vid24 51 frames"
## [1] "Vid25 30 frames"
## [1] "Vid26 50 frames"
## [1] "Vid27 47 frames"
## [1] "Vid28 46 frames"
## [1] "Vid29 40 frames"
```

The mean amount of movement for each roi is calculated (as recorded by the MEA software). If the means of both the therapist and patient are in the top 50 percent, the corresponding data for that roi and time is registered in the list *lista_mov_altos*.

```
lista_mov_altos <- data.frame(matrix(ncol = 15,nrow = 0))

for (v in names(completa)){
    lista_mov_altos <- rbind(lista_mov_altos,
          crea_lista_movs(partition_data(completa[[v]],minutes=d),
                          nombre_video = v,q=quantile,lagmax=lmax))
}

lista_mov_altos<-filter(lista_mov_altos,!leads=="-")

ids <- data.frame(id=1:nrow(lista_mov_altos))
lista_mov_altos<-cbind(ids,lista_mov_altos)

lista_mov_altos$se_mueve_mas = "-"

for (k in 1:nrow(lista_mov_altos)){
  if (lista_mov_altos$mov_patient[k]>1.1*(lista_mov_altos$mov_therapist[k])){
      lista_mov_altos$se_mueve_mas[k]="patient"
  }
```

```r
  if (lista_mov_altos$mov_patient[k]< 0.9*(lista_mov_altos$mov_therapist[k])){
      lista_mov_altos$se_mueve_mas[k]="therapist"
  }
}
```

Adding a column that indicates whether the cross correlations are dominated by positive or negative correlations.

```r
lista_mov_altos$correl<-"-"
x<-which(lista_mov_altos$spearman_max>abs(lista_mov_altos$spearman_min))
y<-which(lista_mov_altos$spearman_max<abs(lista_mov_altos$spearman_min))
lista_mov_altos$correl[x]<-"positive"
lista_mov_altos$correl[y]<-"negative"
MEA_segments<-lista_mov_altos
```

Below, we show the database *MEA_segments* containing 445 segments having high mobility for both subjects. Cross correlations were evaluated and registered for each of these segments. To determine which subject is leading, the sign of the lag with maximum correlation is considered (positive when therapist leads). We show the first few entries, and save the whole database in is saved in the file *MEA_segments.csv* :

```r
kable(head(MEA_segments))
```

id	video	zone	num_period	start	end	mov_patient	mov_therapist	media	spearman_max	lag_max	spearman_min	lag_min	leads	sex	type	se_mueve_mas	correl
1	Vid0	head	12	11.2	12.2	894.9741	130.8486	512.9101	0.0932581	9.10	-0.4102739	-9.93	patient	Male	Treatment	patient	negative
2	Vid0	head	14	13.2	14.2	745.3853	118.0156	431.7004	0.1418647	9.53	-0.4196427	-0.37	patient	Male	Treatment	patient	negative
3	Vid0	head	17	16.2	17.2	755.1611	106.1389	430.6500	0.1739291	13.07	-0.2510538	9.67	therapist	Male	Treatment	patient	negative
4	Vid0	head	18	17.2	18.2	1098.4076	174.4435	636.4256	0.1753803	14.23	-0.3364851	-10.67	patient	Male	Treatment	patient	negative
5	Vid0	head	20	19.2	20.2	698.9201	116.8848	407.9024	0.2114084	10.30	-0.3568905	-6.90	patient	Male	Treatment	patient	negative
6	Vid0	head	29	28.2	29.2	1623.8702	127.9554	875.9129	0.2367328	-6.73	-0.4834532	2.03	therapist	Male	Treatment	patient	negative

```r
write.csv(MEA_segments,"MEA_segments.csv",row.names = FALSE)
```

In the following table we show the frequency in this table for each video and roi.

```r
lista_mov_altos<-group_by(MEA_segments,video,zone)
kable(summarise(lista_mov_altos,frequency=n()))
```

```
## `summarise()` has regrouped the output.
## i Summaries were computed grouped by video and zone.
## i Output is grouped by video.
## i Use `summarise(.groups = "drop_last")` to silence this message.
## i Use `summarise(.by = c(video, zone))` for per-operation grouping
##   (`?dplyr::dplyr_by`) instead.
```

video	zone	frequency
Vid0	head	9
Vid0	legs	5
Vid0	torso	7

video	zone	frequency
Vid12	head	7
Vid12	legs	4
Vid12	torso	5
Vid13	head	9
Vid13	legs	9
Vid13	torso	9
Vid14	head	6
Vid14	legs	6
Vid14	torso	7
Vid15	head	10
Vid15	legs	5
Vid15	torso	7
Vid16	head	8
Vid16	legs	6
Vid16	torso	4
Vid17	head	5
Vid17	legs	5
Vid17	torso	5
Vid18	head	6
Vid18	legs	10
Vid18	torso	8
Vid19	head	8
Vid19	legs	7
Vid19	torso	9
Vid20	head	7
Vid20	legs	12
Vid20	torso	7
Vid21	head	9
Vid21	legs	8
Vid21	torso	6
Vid22	head	6
Vid22	legs	5
Vid22	torso	5
Vid23	head	2
Vid23	legs	8
Vid23	torso	8
Vid24	head	10
Vid24	legs	17
Vid24	torso	10
Vid25	head	8
Vid25	legs	10
Vid25	torso	7
Vid26	head	8
Vid26	legs	17
Vid26	torso	8
Vid27	head	8
Vid27	legs	10
Vid27	torso	10
Vid28	head	11
Vid28	legs	12
Vid28	torso	9
Vid29	head	6

video	zone	frequency
Vid29	legs	6
Vid29	torso	9

Total by video

```
lista_mov_altos<-group_by(MEA_segments,video)
kable(summarise(lista_mov_altos,frequency=n()))
```

video	frequency
Vid0	21
Vid12	16
Vid13	27
Vid14	19
Vid15	22
Vid16	18
Vid17	15
Vid18	24
Vid19	24
Vid20	26
Vid21	23
Vid22	16
Vid23	18
Vid24	37
Vid25	25
Vid26	33
Vid27	28
Vid28	32
Vid29	21

The totals for each roi:

```
lista_mov_altos<-group_by(MEA_segments,zone)
kable(summarise(lista_mov_altos,frequency=n()))
```

zone	frequency
head	143
legs	162
torso	140

And for roi and control/diagnosed groups:

```
lista_mov_altos<-group_by(MEA_segments,sex,type,zone)
kable(summarise(lista_mov_altos,frequency=n()))
```

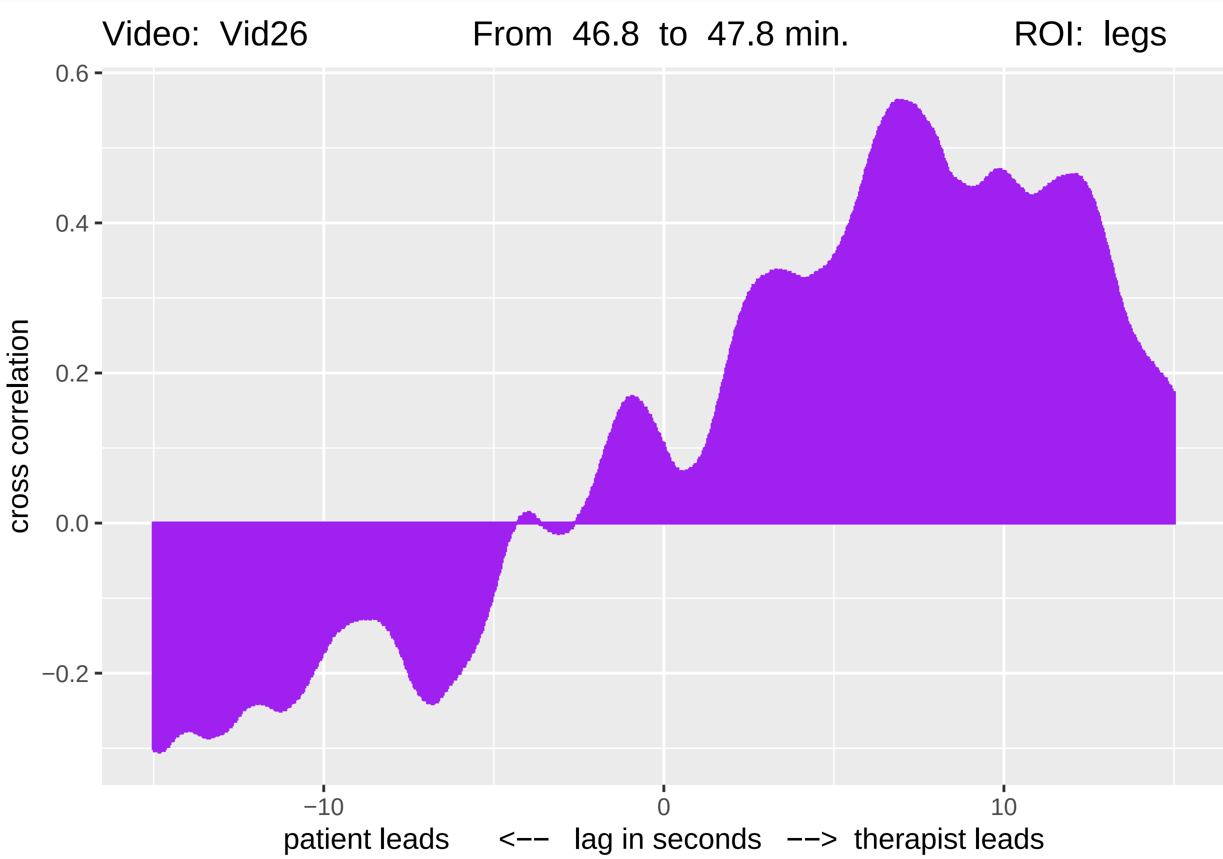
```
## `summarise()` has regrouped the output.
## i Summaries were computed grouped by sex, type, and zone.
## i Output is grouped by sex and type.
## i Use `summarise(.groups = "drop_last")` to silence this message.
## i Use `summarise(.by = c(sex, type, zone))` for per-operation grouping
##   (`?dplyr::dplyr_by`) instead.
```

sex	type	zone	frequency
Female	Control	head	37
Female	Control	legs	38
Female	Control	torso	34
Female	Treatment	head	67
Female	Treatment	legs	82
Female	Treatment	torso	65
Male	Control	head	14
Male	Control	legs	20
Male	Control	torso	15
Male	Treatment	head	25
Male	Treatment	legs	22
Male	Treatment	torso	26

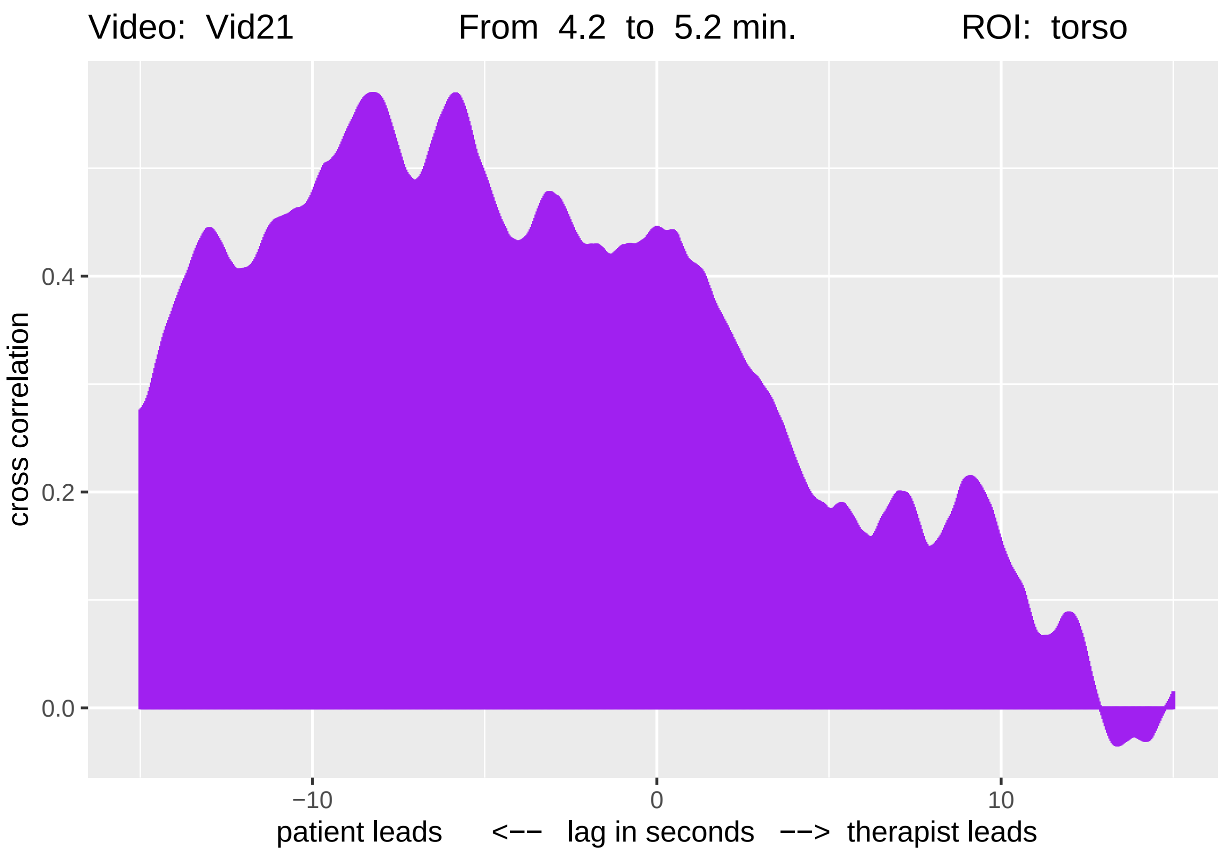
```r
write.csv(summarise(lista_mov_altos,frequency=n()),"Tabla_n2.csv",row.names=FALSE)
```

```
## `summarise()` has regrouped the output.
## i Summaries were computed grouped by sex, type, and zone.
## i Output is grouped by sex and type.
## i Use `summarise(.groups = "drop_last")` to silence this message.
## i Use `summarise(.by = c(sex, type, zone))` for per-operation grouping
##   (`?dplyr::dplyr_by`) instead.
```

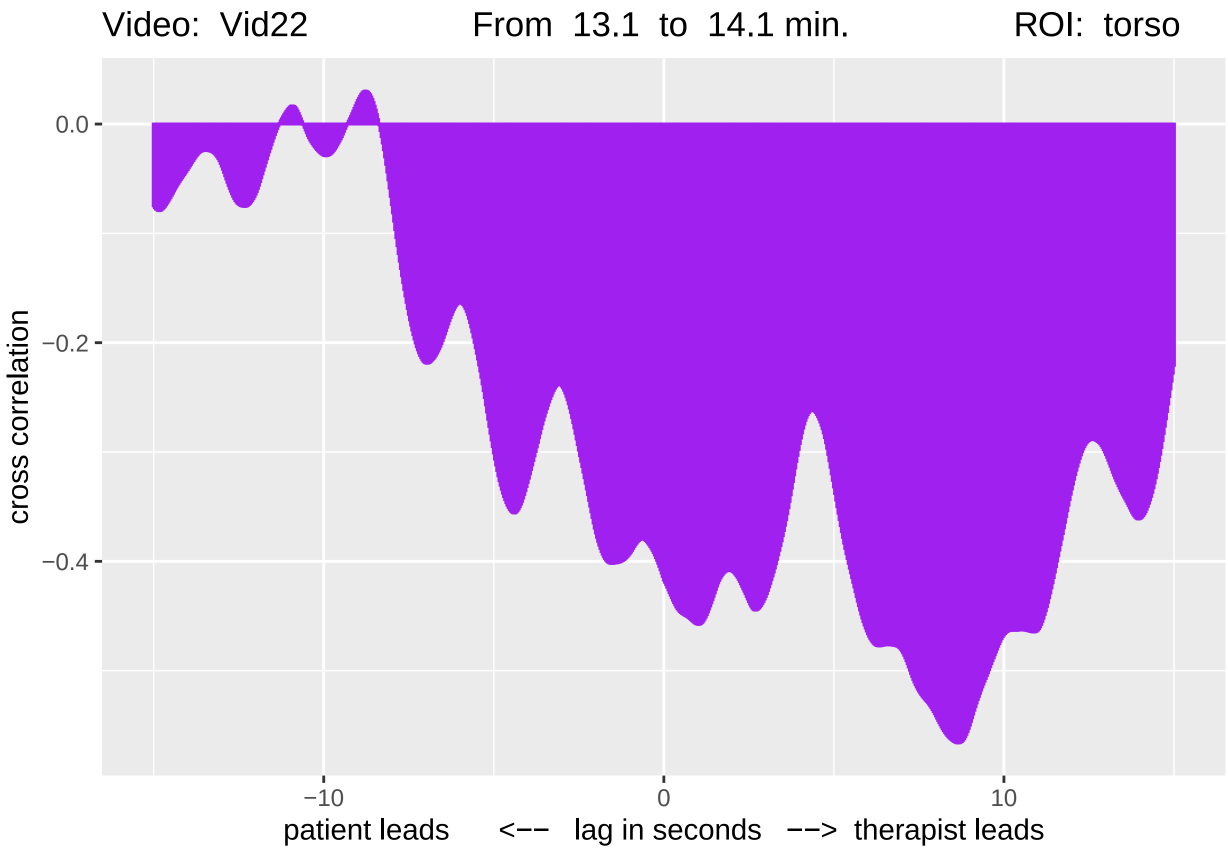
```r
p1<-corr_cruzadas(361,que_lista = lista_mov_altos)
```



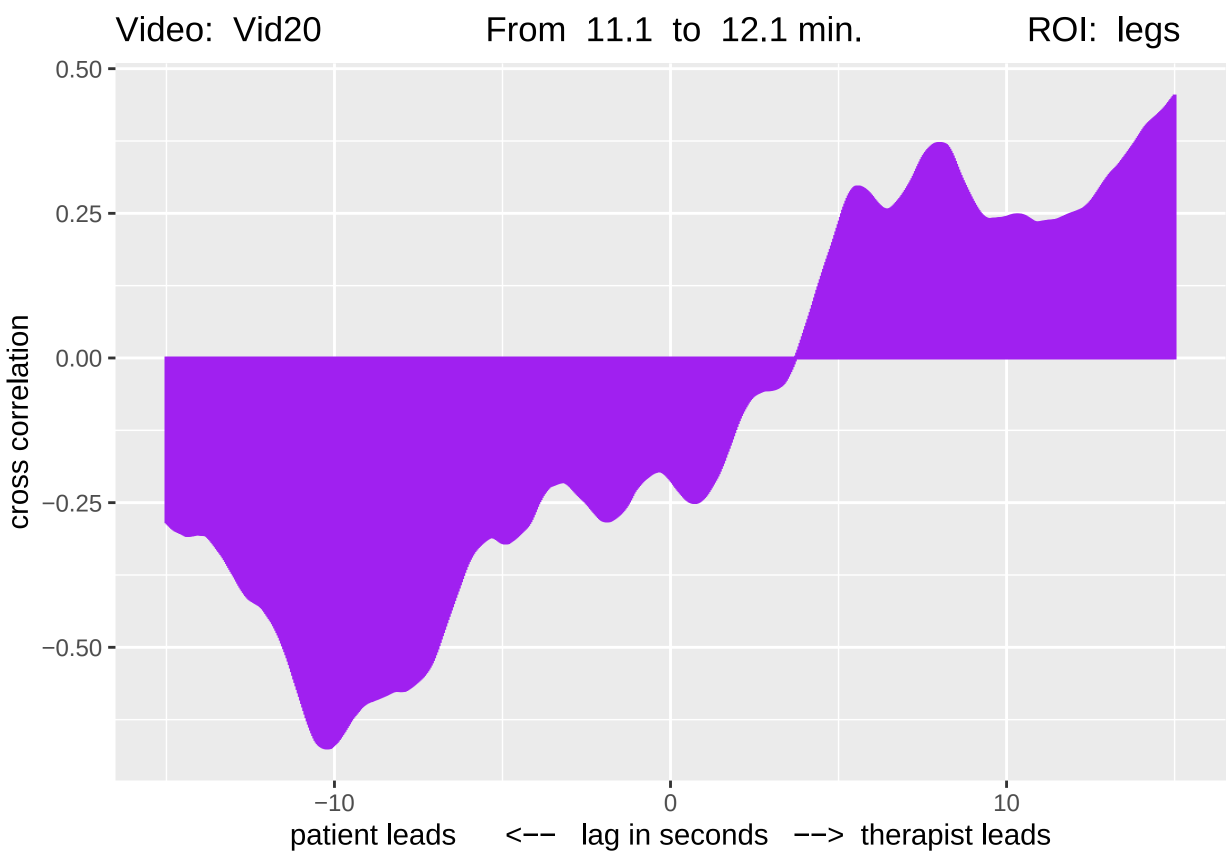
```
p2<-corr_cruzadas(222,que_lista = lista_mov_altos)
```




```
p3<-corr_cruzadas(245,que_lista = lista_mov_altos)
```




```
p4<-corr_cruzadas(203,que_lista = lista_mov_altos)
```




```
png(
  filename = "FigureM1v2_all.png",
  width     = 1950,          # pixels (adjust as you like)
  height    = 3200,
  res       = 200            # dots per inch
)

caption_grob <- textGrob(
  "Figure M1: from top to bottom: dominant positive correlation with therapist lead, dominant positive w
  dominant negative with patient lead, dominant negative with therapist lead.",
  gp = gpar(fontsize=11),
  just = "center"
)

grid.arrange(
  p1, p2, p3, p4,
  nrow = 4,
  top = textGrob(
    "Examples of Cross-correlations (patient vs therapist)",
    gp = gpar(fontsize = 18, fontface = "bold")
  )
)

dev.off()
```

```
## pdf
##   2
```